# Brain

• Know the major parts of the brain and their functions.

• Cerebrum: The largest part, responsible for conscious thoughts and voluntary actions; characterized by the “squiggly” surface.
• Basal ganglia: collection of several brain regions functioning together, not a single structure./ initiate and terminate voluntary body movements
• Limbic system: a group of interconnected brain regions involved in emotional processing and memory. / include the amygdala (central to emotional memory and emotional overlays on memories) and the hippocampus (important in memory formation).

• KNOW ALL OF THE FUNCTIONAL ORGANIZATION…what do they sense? What happens if they are damaged? (eg. primary somatosensory area versus the somatosensory association area…what would the deficit be if the primary somatosensory area was damaged? What would the deficit be if the somatosensory association area was damaged?)

• Diencephalon:
• Thalamus: Major relay station for sensory and motor signals to/from the cortex.
• Hypothalamus: hormone, emotions, eating/drinking, body temperature
• Epithalamus: Pineal gland and habenular nuclei

• Circumventricular organs

**• Brain Stem**
Cerebral Peduncles: Motor tracts transmitting signals from cortex to spinal cord.

Superior Colliculi (visual stimuli): Coordinate head, eye, and trunk movements in response to visual cues.

Inferior Colliculi (auditory stimuli): Orient head and eyes to sudden sounds automatically.

Together, these colliculi form the corpora quadrigemina (four bodies).

Substantia nigra: Key in movement control; produces dopamine. Dysfunction linked to Parkinson’s disease (tremors).

Red nucleus: Also involved in motor coordination.

**• Pons**

Acts as a relay between brain regions.

Plays a critical role in maintaining balance and modulating breathing rate.

Associated with Congenital Central Hypoventilation Syndrome (Ondine's Curse):

A rare disorder where autonomic control of breathing is impaired, requiring mechanical ventilation, especially during sleep.

Can be congenital or caused by brainstem damage.

**• Medulla Oblongata**

Pyramids: Enlargements containing corticospinal tracts that decussate (cross over), explaining why each brain hemisphere controls the opposite body side.

Nuclei clusters perform vital autonomic functions:

Cardiovascular control (heart rate regulation)

Respiratory control (basic breathing rhythm)

Vomiting center: Initiates vomiting reflex, which can occur unconsciously, posing risks in concussions or intoxication.

Deglutition: Controls swallowing reflex.

**• Cerebellum**

Located posterior to brainstem.

Coordinates skeletal muscle contractions, regulates posture and balance.

Involved in motor learning (e.g., riding a bike, typing).

Plays a role in cognition, memory, and language processing, though not the primary language center.

Alcohol impairs cerebellum function, causing impaired balance and coordination.

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# Know the meninges:

Dura mater: The tough, outermost layer; very thick and durable.

Arachnoid mater: A middle layer named for its spider-web-like appearance; contains the arachnoid space.

Pia mater: The thin, innermost layer adherent to the brain surface.

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• What is the purpose of the blood brain barrier? What does it consist of? What can cross it easily, via active transport or not at all?

Protects the brain from harmful substances and pathogens

Consists of tight jnx of epithelial cells and the permeability is maintained by astrocytes

Things that cross the BBB:

– Glucose via active transport

– Creatinine, urea, ions – slowly

– Proteins and most antibiotic drugs – no passage

– Lipid soluble substances ($O_2$, $CO_2$, alcohol, anesthetics)

– easily cross

• What is the function of pineal gland?

Secretes melatonin

• What are the functions of the cerebral spinal fluid? Where is it made? Diagram the flow.

• Know the difference between the tracts in the cerebrum

| Tract Type | Description | Example/Function |
|---|---|---|
| **Association tracts** | Connect gyri within the same hemisphere | Front-to-back connections on one side |
| **Commissural tracts** | Connect gyri across hemispheres (left-right) | Largest example: **Corpus Callosum** |
| **Projection tracts** | Connect cortex with other brain regions (e.g., brainstem) | Carry signals between cortex and lower CNS |

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# What are:

Capgras Delusion: Patients recognize familiar people visually but lack emotional response, leading to the belief that these individuals are impostors.

Fregoli Syndrome: Patients believe a single person is disguising themselves as multiple different people, due to misinterpretation of emotional cues.

Cotard Delusion: Patients fail to emotionally recognize themselves, believing they are dead or non-existent; often leads to severe depression and suicidal tendencies.

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## What are the symptoms of:

| Aphasia Type | Location of Damage | Key Deficits | Speech Production | Comprehension | Repetition |
|---|---|---|---|---|---|
| Broca's Aphasia | Broca's area (motor speech area) | Difficulty producing speech; impaired complex sentence comprehension | Non-fluent, effortful | Good for simple sentences | Variable |
| Wernicke's Aphasia | Wernicke's area (language comprehension) | Fluent but nonsensical speech; poor comprehension | Fluent but meaningless | Poor | Poor |
| Conduction Aphasia | Connection between Broca's & Wernicke's areas | Intact comprehension and speech production; impaired repetition | Fluent, meaningful | Good | Poor |
| Global Aphasia | Broca's & Wernicke's areas + connections | Severe deficits in all language functions | Severely impaired | Severely impaired | Severely impaired |

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# Autonomic Nervous System AND Sensory, Motor, and Integrative Systems

• What are the functions of the parasympathetic and sympathetic nervous system? Know the details about each.

Sympathetic: fight or flight / pupils dilate, sweating, heart rate increased,

Para: Dominant during rest / digestion, decrease in heart rate, etc

• Difference between endoneurium, perineurium, and epineurium

**• Difference between a sensation and a perception**

Sensation: The conscious or subconscious awareness of environmental changes (internal or external). Example: sensing a change in light intensity.

Perception: The conscious interpretation and meaning assigned to a sensation. Example: interpreting bright light as unpleasant or darkness as scary.

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## What do the following do:

Exteroceptors: Detect stimuli from the external environment (e.g., touch, temperature).

Interoceptors: Detect stimuli from internal organs (e.g., stretch receptors in the stomach).

| Receptor Type | Stimulus Detected | Examples |
|---|---|---|
| **Proprioceptors** | Body position and movement | Joint and muscle sensors |
| **Mechanoreceptors** | Physical stimuli like touch and stretch | Skin receptors, stomach stretch sensors |
| **Thermoreceptors** | Temperature changes | Skin temperature sensors |
| **Nociceptors** | Pain stimuli | Injury-related pain receptors |
| **Photoreceptors** | Light (vision) | Retina cells |
| **Chemoreceptors** | Chemical stimuli | Smell, taste, blood pH sensors |
| **Osmoreceptors** | Hydration levels | Body fluid balance sensors |

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- **Rapidly Adapting (Phasic) Receptors:**
  - Respond quickly at the onset and offset of a stimulus but not during sustained stimulation.
  - Examples: smell, pressure, and touch.
  - Function: Detect changes or new stimuli but ignore constant, unchanging stimuli (e.g., smell of a room fades after a few minutes).
- **Slowly Adapting (Tonic) Receptors:**
  - Continue to respond throughout the duration of the stimulus.
  - Examples: pain and proprioception.
  - Function: Provide continuous information about stimulus presence, important for safety and body awareness (e.g., persistent pain signals injury).

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## What do the following do:

| Receptor Name | Location | Function | Adaptation Type | Notes |
|---|---|---|---|---|
| **Free nerve endings** | Epidermis (superficial) | Pain detection | *Not specified* | Example: nose pain receptors |
| **Merkel discs** | Epidermis-Dermis border | Discriminative touch (fine texture) | Slowly adapting | 25% of hand receptors; highly sensitive |
| **Meissner's corpuscles** | Dermal papillae | Discriminative touch & vibration | Rapidly adapting | Detect initial touch, then adapt quickly |
| **Hair root plexus** | Around hair follicle | Detects hair movement | Rapidly adapting | Senses movement like a bug touching hair |
| **Ruffini corpuscles** | Mid-dermis | Heavy touch and pressure | Slowly adapting | Detects sustained pressure, less sensitive to fine details |
| **Pacinian corpuscles** | Deep dermis/subcutaneous | Pressure and high-frequency vibration | Rapidly adapting | Encapsulated; detects deep pressure and vibration |

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• How do you sense temperature?

Through thermal sensations

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# Know the different types of pain:

**Types of Pain:**

Fast pain: Immediate, sharp pain (e.g., cutting or hitting oneself).

Slow pain: Dull, aching pain that develops over time (e.g., delayed onset muscle soreness or injuries from accidents).

**Pain Location Classification:**

Superficial somatic pain: Occurs in skin or surface tissues (e.g., light bruises).

Deep somatic pain: Affects muscles, bones, or deep tissues (e.g., deep aches after injury).

Visceral pain: Originates from internal organs (heart, lungs, intestines) and can be diffuse or radiating.

**Diffuse Pain**: Pain spreads beyond the site of injury (e.g., kidney pain radiating to thigh areas).

**Referred Pain**: Pain perceived in a location distant from the source due to shared spinal cord innervation.

Classic example: Heart attack pain felt down the left arm, despite no injury there.

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• Name the spinal nerve pairs and state how many of each there are.

| Region | Number of Nerve Pairs | Number of Vertebrae | Notes |
|---|---|---|---|
| Cervical | 8 | 7 | Extra nerve exits above atlas |
| Thoracic | 12 | 12 | Equal number |
| Lumbar | 5 | 5 | Equal number |
| Sacral | 5 | 5 | Equal number |
| Coccygeal | 1 | 1 (coccyx) | Single nerve pair |

**• What is the difference between the dorsal root and the ventral root? What does the dorsal root ganglion contain?**

Dorsal: incoming sensory fibers

Ventral: Outgoing motor fibers

Dorsal root ganglion: cell bodies of sensory fibers

• What is the difference between a nerve and a tract?

**• What do the gray matter and white matter do?**

Gray matter is the shaped like the letter H or a butterfly.

- Receives and integrates incoming and outgoing information.

White matter is divided into columns.

- Transmits nerve impulses to and from the brain

**• In spinal tracts, what is meant by a first order, second order and third order neuron?**

First: receptor to brain stem/spinal cord

Second: Brain stem to spinal cord to thalamus

Third: Thalamus to primary somatosensory area of cortex

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• **What is a reflex?**

A fast, predictable, automatic response to environmental changes that helps maintain homeostasis.

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## Know what is happening in the:

| Reflex Type | Description | Key Mechanism |
|---|---|---|
| **Stretch Reflex** | Muscle contraction in response to stretching (e.g., knee-jerk reflex) | Muscle spindles detect stretch → signal spinal cord → agonist contracts, antagonist relaxes |
| **Tendon Reflex** | Protects muscle from excessive tension by inhibiting contraction | Tendon organs sense tension → inhibit agonist muscle contraction → antagonist contracts |
| **Flexor Withdrawal Reflex** | Withdrawal of limb from painful stimulus (e.g., stepping on a tack) | Sensory input to spinal cord → multiple muscles activated to lift limb away from stimulus |
| **Crossed Extensor Reflex** | Balances body by activating muscles on opposite side when one limb withdraws (e.g., maintaining balance) | Coordinated with flexor withdrawal → opposite limb muscles contract to support body weight |

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**Basal Ganglia**: Group of brain regions important for initiating and terminating movements.

Damage leads to movement disorders such as Parkinson's disease, characterized by abnormal involuntary movements.

**Cerebellum**: Monitors intended movement and actual movement, providing feedback for coordination and balance.

Example: When intoxicated, cerebellar function is impaired, causing uncoordinated movement (e.g., inability to walk a straight line).

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• What is the reticular activating system? What does it do? What triggers it? What doesn't?

Network of brain regions that regulate wakefulness and arousal.

Activated by stimuli such as pain, light, noise, and touch (but not smell).

Plays critical role in maintaining consciousness.

Explains why some stimuli (e.g., alarm clocks) wake us, but smell alone does not.

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## Know the areas that the following supply:

• Cervical: Supplies skin and muscles of the head, neck, and upper shoulders.
• Brachial: Innervates the shoulder and upper limb.

• Lumbar: Supplies the anterior thigh and lower abdomen.
• Sacral: Includes the sciatic nerve, a large nerve supplying lower limbs.

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## What are: What causes them?

• Neuritis: Inflammation of nerves caused by injury, vitamin deficiency, toxins, or infections.
• Polio: Viral infection that destroys motor neurons.
• Shingles: Reactivation of the chickenpox (varicella-zoster) virus.

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# Special Senses

**• Know the different cells of the olfactory membrane**

Olfactory receptors – bipolar neurons with cilia or olfactory hairs

• Supporting cells – columnar epithelium

• Basal cells = stem cells – replace receptors monthly

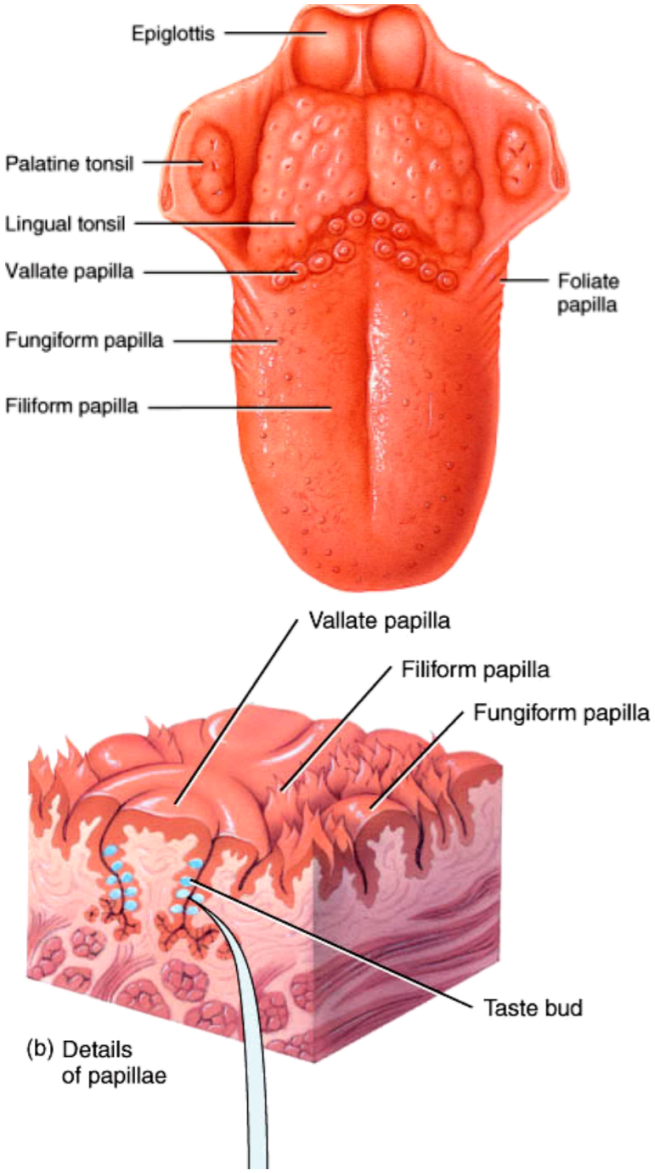
• Olfactory glands - produce mucus

**• What is the vomeronasal organ? What does it detect?**

Present and functional in many animals for pheromone detection.

## What are the different senses of taste?

• Where are your taste buds located? Know the different papillae on the tongue.




tongue, soft palate & larynx

• Know the physiology of taste: salty, sour, bitter, umami and sweet

## Know all of the accessory structures of the eye and what they do:

Eyelids (Palpebrae) protect the eye and spread tears to lubricate the surface.

Tarsal glands (Meibomian glands) secrete oils preventing eyelids from sticking.

Conjunctiva is a transparent mucous membrane with two parts:

Palpebral conjunctiva (lining eyelids)

Bulbar conjunctiva (covering the eyeball)

Conjunctiva contains blood vessels; redness (bloodshot eyes) occurs here, not in the sclera.
- Eyelash
- Eyebrow
- Sebaceous glands
- Palpebral fissure
- Lacrimal caruncle

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• Know the different parts of the lacrimal apparatus. Where do tears drain into?

- Produces ~1 mL of tears per day via the lacrimal gland (located at the superior-lateral eye).
- Tears lubricate and have antibacterial enzymes.
- Tears drain through lacrimal canals into the nasolacrimal duct, emptying into the nose, explaining why crying causes a runny nose.
- Lacrimal caruncle, the pink fleshy area in the eye corner, produces oily secretions that can crust (white goop).

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## Know the different tunics of the eye, what they do and what is contained in each:

| Layer (Tunics) | Components | Function/Notes |
|---|---|---|
| **Fibrous Tunic** | Sclera (white part), Cornea (clear) | Protection; cornea refracts light; sclera is dense connective tissue |
| **Vascular Tunic** | Iris, Pupil, Choroid, Ciliary body | Blood supply; controls light entry; focuses lens; absorbs stray light |
| **Nervous Tunic** | Retina (photoreceptors, optic disk) | Contains neurons/receptors; optic disk = blind spot |

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• Where is the anterior and posterior cavity of the eye located? What is in each?

| Cavity | Fluid Type | Characteristics | Notes |
|---|---|---|---|
| Anterior cavity | Aqueous humor | Clear, watery, produced by ciliary body | Replaced every ~90 mins; drains through canal of Schlemm |
| Posterior cavity | Vitreous humor | Jelly-like, formed once, not replaced | Floaters are debris in this gel |

• **What is glaucoma? What causes it?**

Glaucoma results from impaired aqueous humor drainage, increasing intraocular pressure and potentially damaging optic neurons.

• **What is the tapeta lucida?** is a reflective layer in nocturnal animals enhancing night vision by reflecting light back onto photoreceptors, causing eye shine but reducing sharpness.

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# What are:

Myopia (nearsightedness): Eye too long; distant objects blurred.

Hyperopia (farsightedness): Eye too short; near objects blurred.

Astigmatism: Irregular corneal or lens curvature causing distorted images.

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• What are the functions of rods and cones?

• What is the macula lutea? Fovea centralis?

• What is meant by the convergence of the eyes?

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# What structures are located in the:

**External Ear**

Known as the auricle (or pinna), made of elastic cartilage covered with skin.

Its primary function is to collect sound waves. Animals with large ears can orient them toward sound sources to improve hearing; humans can mimic this by cupping their hands around their ears.

The external auditory canal (~1 inch long) channels sound waves inward. It contains glands that produce cerumen (earwax) for protection.

The canal ends at the tympanic membrane (eardrum), which vibrates in response to sound.

**Middle Ear**

An air-filled cavity containing three small bones called the ossicles: malleus, incus, and stapes.

The stapes footplate rests on the oval window, transmitting vibrations into the inner ear fluids.

The round window acts as a flexible membrane that bulges to accommodate fluid movement, allowing pressure equalization.

Contains the auditory (Eustachian) tube, which connects to the throat to equalize middle ear pressure (e.g., when ears “pop” during altitude changes).

Two tiny muscles—stapedius (attached to stapes) and tensor tympani (attached to malleus)—help regulate ossicle movement, protecting the ear from chronic loud noises by stiffening the ossicular chain.

**Inner Ear (Labyrinth)**

Consists of the bony labyrinth (structural bone) and the membranous labyrinth (fluid-filled tubes inside the bony labyrinth).

The bony labyrinth includes the semicircular canals and the cochlea (snail shell-shaped structure).

The semicircular canals and vestibule are responsible for equilibrium and balance.

The cochlea is the primary organ of hearing.

The vestibulocochlear nerve carries signals from both balance and hearing organs to the brain.

The fluids in the inner ear are perilymph (in bony labyrinth) and endolymph (in membranous labyrinth).

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• What is the difference between nerve deafness and conduction deafness?

| Type | Cause | Affected Structures | Example Causes |
|---|---|---|---|
| **Nerve Deafness** | Damage to vestibulocochlear nerve or hair cells | Nerve fibers or cochlear hair cells | Antibiotics, loud high-pitched sounds, anti-cancer drugs |
| **Conduction Deafness** | Sound vibrations fail to reach cochlea | Eardrum, ossicles (middle ear bones) | Perforated eardrum, fused ossicles |

• What do the semicircular canals do?

Fluid in semicircular canal ducts moves during head rotations.

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• What is the difference between static and dynamic equilibrium? How is each detected? What structures are involved?

**Static Equilibrium**

Maintains awareness of head and body position when stationary.

Uses calcium carbonate crystals (“otoliths”) embedded in a gelatinous membrane atop hair cells.

Gravity causes these crystals to shift when the head tilts, bending hair cells.

Bending of hair cells opens ion channels, causing depolarization and sending signals to the brain about head orientation.

**Dynamic Equilibrium**

Maintains balance during movement (e.g., acceleration, turns).

Fluid in semicircular canal ducts moves during head rotations.

Movement displaces the cupula (gelatinous dome) in the ampulla, bending hair cells.

This bending opens ion channels, triggering neural signals interpreted by the brain as motion and spatial orientation.